

André Cédric Bessala, Ph.D. Research Scientist

IRIT Laboratory/Toulouse-INP, University of Toulouse, France, andrecedric.bessala@toulouse-inp.fr

RESEARCH EXPERIENCE

PhD Research, IRIT Laboratory/Toulouse-INP, University of Toulouse (2023-present)

- Investigating advanced techniques for enhancing efficiency and flexibility in NextG Mobile Systems (5G & B5G)
- Designing and evaluating Open Radio Access Network (O-RAN) slicing architectures for dynamic and scalable network services
- Applying intent-based networking principles to automate and optimize network orchestration
- Developing single- and multi-agent reinforcement learning (RL) frameworks for adaptive orchestration in highly dynamic environments with inconsistent data availability
- Optimizing energy efficiency and performance trade-offs through intelligent resource allocation
- Modeling and simulating large-scale O-RAN scenarios to evaluate QoS guarantees, latency constraints, and energy savings under real-world traffic patterns

Master Research, Department of Computer Science, University of Yaoundé I (2022-2023)

- Developing real-time traffic forecasting models for software-defined networking (SDN) environments
- Designing and implementing network monitoring patterns for performance analysis and anomaly detection
- Applying machine learning techniques and classification algorithms to address network management challenges
- Utilizing ensemble learning methods (XGBoost, CatBoost, Random Forest) for improved prediction accuracy and robustness
- Evaluating system performance under varying traffic loads to optimize scalability and reliability

PERSONAL SKILLS

- Mother tongue: French
- Other language English: Fluent
- Communication skill:
 - Good communication skills acquired by exposing research activities in meetings and workshops
 - Good brainstorming skills
- Organisational / managerial skills:
 - Good work organization, subdivision and rationalization
 - Good team working and leading skills achieved during the time spent working as tutor for undergraduate student
- Computer skills:
 - Programming Languages: Python, Java, C++, C, Latex, Matlab, MySQL, HTML, Javascript, CSS, UML
 - Engineering Tools and Software: Pytorch, PyOMO, Matlab, Cplex, Gurobi, WireShark, Mininet, SLICES-RI, Grid'5000, Colosseum

- Operating Systems: Linux/Unix, Windows
- Technical skills:
 - Networking: Wired and wireless network design, configuration, and performance optimization
 - Optimization & Modeling: Game theory, resource allocation strategies, forwarding/routing/relaying policies
 - Artificial Intelligence: Machine learning, reinforcement learning, multi-agent systems for network control
 - Energy Efficiency: Energy-saving algorithms and performance-energy trade-off optimization in communication systems
 - Security & Privacy: Threat mitigation, secure communication protocols, privacy-preserving mechanisms
 - Experimentation: Network simulation, emulation, and real-world testbed deployment
 - Software Engineering: Design, development, and testing of software solutions for networked systems

EDUCATION

University of Toulouse (2023-2026)

PhD Computer Science and Telecommunications: Adaptive Energy-Efficient RAN Function Placement in O-RAN

University of Yaoundé I (2022-2023)

MS Degree Computer Science: Intelligent Traffic Qualification in a Programmable Wireless Mesh Network

TEACHING EXPERIENCE

Mentor/Supervisor, Toulouse INP-ENSEEIH, Department of Numeric Sciences (2025-2026)

I intervene in different teaching units related to Networks and Computer Science in general as a Mentor/Supervisor. Those teachings span across all the 3 levels (L3, M1, M2) of Toulouse INP-ENSEEIH core engineering cycle and also cover Bachelor.

M2: Master 2, M1: Master 1, L3: License 3 (Bachelor level), 64 hours completed.

- Delivering practical and tutorial sessions for courses: Internet, Local Networks, Quality of Service, Mobiles Programming, Infrastructure of Big Data, Infrastructure of Clouds.

Mentor/Supervisor, Toulouse INP-ENSEEIH, Department of Numeric Sciences (2024-2025)

I intervene in different teaching units related to Networks and Computer Science in general as a Mentor/Supervisor. Those teachings span across all the 3 levels (L3, M1, M2) of Toulouse INP-ENSEEIH core engineering cycle and also cover Bachelor.

M2: Master 2, M1: Master 1, L3: License 3 (Bachelor level), 64 hours completed.

- Delivering practical and tutorial sessions for courses: Internet, Local Networks, Quality of Service, Mobiles Programming, Infrastructure of Big Data, Infrastructure of Clouds.

Mentor/Supervisor, Toulouse INP-ENSEEIH, Department of Numeric Sciences (2023-2024)

I intervene in different teaching units related to Networks and Computer Science in general as a Mentor/Supervisor. Those teachings span across all the 3 levels (L3, M1, M2) of Toulouse INP-ENSEEIH core engineering cycle and also cover Bachelor.

M2: Master 2, M1: Master 1, L3: License 3 (Bachelor level), 64 hours completed.

- Delivering practical and tutorial sessions for courses: Internet, Local Networks, Quality of Service, Mobiles Programming, Infrastructure of Big Data, Infrastructure of Clouds.

REFEREES

Prof Gentian Jakllari, Toulouse INP , IRIT-CNRS, gentian.jakllari@toulouse-inp.fr

Prof Guilherme Iecker Ricardo, Toulouse INP, IRIT-CNRS, guilherme.ieckerricardo@toulouse-inp.fr

PUBLICATIONS

E-MATRO: An Adaptive and Energy-Efficient Framework for Slicing within the O-RAN Architecture

André Cédric Bessala , Guilherme Iecker Ricardo , Gentian Jakllari

24th IFIP Networking Conference (NETWORKING 2025), International Federation for Information Processing (IFIP); IEEE, May 2025, LIMASSOL, Cyprus

POSTERS

Adaptive Energy-Efficient RAN Function Placement in O-RAN

PEPR FUTURE NETWORKS, NF-NAI PROJET, Trans Numériques, Rennes, 2-5 mars 2026

E-MATRO: An Adaptive and Energy-Efficient Framework for Slicing within the O-RAN Architecture

PEPR FUTURE NETWORKS, NF-NAI PROJET, Bordeaux, 2025 June 01/03

Energy-Efficient Traffic Steering for 5G and Beyond Slicing,

PEPR FUTURE NETWORKS, NF-FAI PROJET, Grenoble, 2024 March 22/24

TALKS

IRIT-CNRS PhD Event Day, Toulouse, 2025 August 08, Adaptive Energy Saving within the ORAN Framework via MARL

24th IFIP Networking Conference (NETWORKING 2025), 28 May 2025, LIMASSOL, Cyprus, E-MATRO